Poor isometric neck extension strength as a risk factor for concussion in male professional Rugby Union players

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ABSTRACT

Background Concussion is one of the highest burden injuries within professional Rugby Union ('rugby') and comes with a high health and financial cost to players and teams. Limited evidence exists as to the existence of modifiable intrinsic risk factors for concussion, leaving athletes and clinicians with few options when developing prevention strategies.

Objective To investigate whether neck strength is significantly associated with concussion incidence in professional male rugby players.

Methods 225 rugby players were assessed for neck strength at three time points throughout the 2018/2019 season using a method of isometric contraction. Associations with clinically diagnosed concussion injuries are presented as incidence rate ratios (IRRs) with 95% CIs.

Results Thirty concussions occurred in 29 players during the study period; a rate of 13.7 concussions per 1000 hours played. Greater neck strength was observed at mid and end of season time points versus preseason across the study population. There was a significant association between extension strength and concussion; a 10% increase for extension strength was associated with a 13% reduction in concussion rate (adjusted IRR (95% CI) 0.87 (0.78 to 0.98). No other significant associations were observed between concussion incidence and any other unique neck strength range or composite score.

Conclusion Higher neck extension strength is associated with lower concussion rates in male rugby players. Neck strength is a modifiable intrinsic risk factor for concussion and may be an important component of a strength and conditioning regime.

INTRODUCTION

Concussion is a microstructural and functional injury of the brain, resulting from external biomechanical forces, usually from an opposition player or as a result of technique error.[1 2] Symptoms are usually rapid in onset and can last from minutes to months,[3] with no current ability to accurately predict length or severity of presentation.[4] In professional Rugby Union ('rugby'') and within contact sports at large, sports-related concussion is becoming one of the highest burden injuries when considering frequency and time missed from competition.[5 6] While the mechanism of injury that precedes a concussive event is variable and often position specific,[7] research demonstrates that concussions are most commonly associated with forces, resulting in linear or rotational movements of the head and neck.[8] Within rugby, 58%–64% of all match concussions in professional male players occur during the tackle[9 10]; the most common concussive event, followed by the ruck, kicking contests and mauls.[11] When engaging in a contact event during rugby, acceleration of the head has been found to occur within the sagittal plane in 50% of concussive events,[12] suggesting a link with the flexor and extensor mechanism of the head and neck.

Laboratory and field-based studies have provided indications as to the mechanism and potential mitigating factors of a concussive injury.[13–15] Muscle function is thought to be a significant factor in attenuating an external impact force and, therefore, the head acceleration(s) thought to lead to concussive injury.[16] Evidence is conflicting, however, and although muscle function contributes approximately 80% of the minimally needed mechanical stability of the cervical spine,[17] research comparing a player's isometric muscle strength, muscle size and response to cervical perturbations against head impact biomechanics has found no mitigating impact of increased neck strength and in some cases, observed an increase in impact severity.[13 15] In contrast, earlier work has demonstrated that greater neck strength and anticipatory cervical muscle activation can reduce the magnitude of the head's kinematic response.[14] This is supported by the sole field-based study investigating the association between neck strength and concussion incidence. Collins *et al*[18] observed neck strength in a group of 6704 mixed sport and gender high school athletes and concluded that low neck strength, when averaged across four ranges, was a significant risk factor for concussion.[18] There were, however, no Rugby Union players in this cohort and no published evidence currently exists considering this sporting population and the impact of neck strength on concussion incidence.

Higher velocity contact scenarios in rugby union lead to an increased risk of concussion.[19–21] It is possible that activation of the neck and posterior shoulder muscles may reduce the risk of concussion by stabilising the head and neck during impact, thereby decreasing the resultant head acceleration.[22] Within rugby, the largest isometric neck forces generated by professional male players exist in the extension range, approximately 90% higher than the peak lateral flexion values.[23] Forwards generate significantly larger neck strength forces

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than backs[23–26] across test parameters, which may be explained by their usually greater neck girth and length.[23] When comparing concussion incidence between positions, the evidence for neck strength helping to mitigate the risk of concussion is contradictory but demonstrates consistently higher rates of concussion in backs.[9 27] Neck strength is, however, only one variable in this analysis and differences in the physical requirements of different rugby positions must be considered.

The 2016 Concussion in Sport Group consensus statement states that a clear understanding of the potentially modifiable risk factors required to design, implement and evaluate appropriate injury risk reduction strategies is needed.[16] Therefore, we aimed to establish if reduced neck strength was a risk factor for concussion in professional male rugby players. We hypothesise that reduced neck strength will be positively associated with an increased concussion incidence in this population.

METHODS

Patient and public involvement

No patient or public involved.

Participants

Following ethical approval by University College London (11785/004), participants for this prospective cohort study were recruited from 10 professional rugby teams competing in the Georgian 'Big 10' league, the highest level of competition in Georgia. Of 225 participants aged between 18 and 35 years consented to participate following a communique sent by their National Governing Body. To be included, participants had to demonstrate no evidence of pain through their cervical or glenohumeral range of movement following assessment by one of two clinicians (a physiotherapist and osteopath). Participants were assessed for neck strength at preseason, mid-season (week 16) and end of season (week 27, post playoffs) time points and monitored for concussion incidence over the course of a full season, including playoff matches.

Concussion and match exposure data

A clinical diagnosis of concussion was made by team doctors using guidelines set out by the Consensus Statement on Concussion in Sport, following the fifth International Conference on Concussion in Sport.[16] They were blinded to the results of our neck strength assessment and were experienced in the diagnosis and management of concussion. All club doctors had all completed the World Rugby concussion education modules and a preseason concussion identification workshop run by the Georgian Rugby Union. Following a confirmed diagnosis, the club medical department notified the research team via email or phone call, providing the name of the concussed player and the date of injury. Number of games played and match minutes were used to calculate players' match exposure. Exposure data were collected from the Georgian Rugby Football Union via the league recording programme.

Testing protocol

A testing protocol described and validated by Versteegh *et al*[28] was employed to assess neck strength. Participants were seated on a treatment bed facing a mirror with their feet firmly grounded. No back or arm support was offered in order to prevent bracing. One of two clinicians then guided the participant through the testing procedure.

Isometric neck strength was tested in six ranges using a hand-held dynamometer (HHD) (Lafayette digital hand-held dynamometer) to evaluate maximum force generated in kilogram-force. Assessment was executed in the order of: forward flexion (with resistance applied with two hands to the forehead), extension (with resistance applied with two hands to the occiput), right and left side flexion (with resistance applied with the ipsilateral hand above the ear) and right and left rotation (with resistance applied along the mid jaw with the ipsilateral hand). For rotation and side flexion ranges, the shoulder was abducted to 90° and the participant was instructed to keep the elbow high using their reflection in the mirror as a reference.

Participants were instructed to build up to a maximal effort over 3 s and stop when they heard the HHD 'beep', indicating that maximal effort had been achieved. Each range was assessed consecutively without rest in the order described above. Following the completion of each round, a 1 min rest was provided before the next round commenced in the same order. Three rounds of assessment were undertaken with the participants' mean score was used for analysis.

Prior to the assessment of maximal testing, two warm-up protocols were executed. A horizontal adduction test was undertaken to ensure that upper limb strength was adequate to overcome the force generated by the neck musculature and a round of submaximal neck muscle warm up efforts were undertaken using the methods described above. Each participant was instructed to apply 50% of their maximal pressure in a 5 s isometric contraction across each range (figure 1).

Inter-rater reliability testing

Using the methods described above, 15 participants were assessed by both raters. Following the initial assessment by rater one, subjects were given a 10 min rest and then asked to return to the assessment area to be assessed by rater two. The same assessment order was followed by each rater; forward flexion, extension, right side flexion, left side flexion, right rotation and left rotation. Rater 2 was blinded to rater 1 results and the order that each rater-assessed participants was chosen at random.

Statistical analysis

Data were analysed using Stata V.14 (StataCorp, Texas). The study was powered to detect an effect size of 0.80 for a 10% increase in total neck strength with 80% power at the 5% significance level (calculated using GPower V.3.1). Normality was assessed using histograms and quantile–quantile plots.

Demographic variables are presented by concussion status as mean (SD) for normally distributed data, median (IQR) for non-normal and % (N) for categorical data and compared using two-sample t tests, Mann-Whitney U test and χ^2 tests, respectively. Since distributions of the strength variables were skewed, log transformation was used to meet the model assumptions. Geometric means and geometric SDs were obtained by exponentiating the means and SDs on the log scale. Changes in strength over time were modelled using a linear mixed model.

A random intercept was fitted for player identification, and time was fitted as a fixed effect using two dummy variables to allow mid and postseason to be compared with preseason as the reference category. The model was then reparameterised to allow comparison of the mid and preseason values. Percent differences and 95% CIs are presented. Mean values were plotted on the log scale to show the changes over time. The significance level for the pairwise comparisons was adjusted for multiple comparisons using the Bonferroni correction.

Concussion incidence and strength were analysed by fitting a Poisson regression model with number of hours played as the

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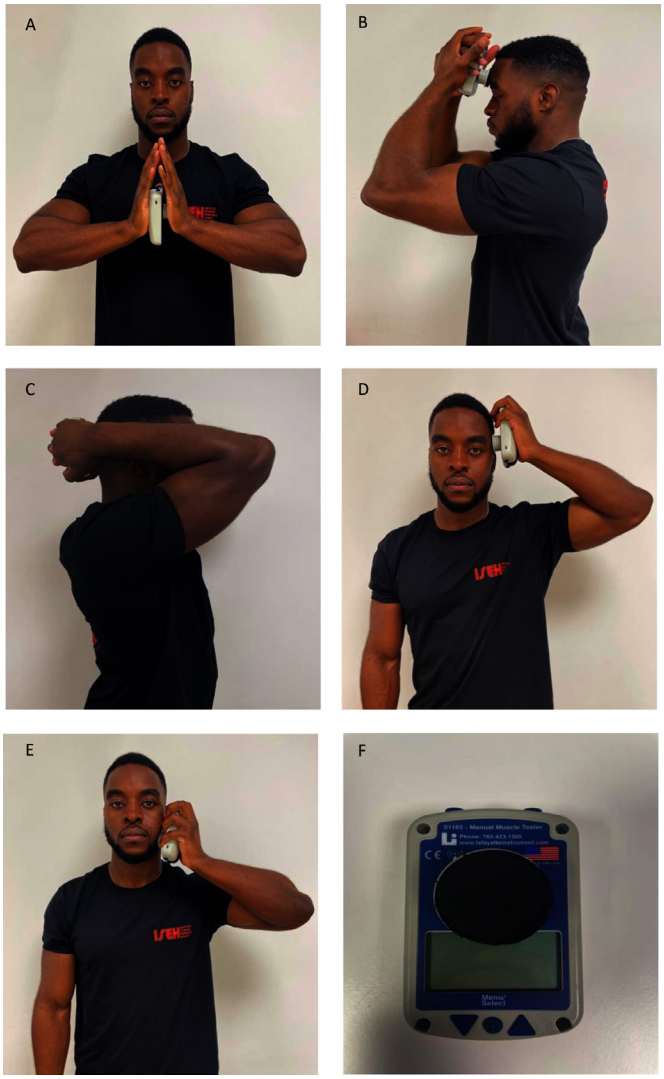

Figure 1 Test positions: HADD test (A),forward flexion (B),extension (C),side flexion (D),rotation (E) and handheld dynamometer (F). HADD, horizontal adduction.

exposure variable. A priori covariates included were age, club and body mass index (BMI).

Associations of neck strength with concussion rate outcome are presented as incidence rate ratios (IRRs) with 95% CIs. As neck strength was log transformed in the analysis, we present the IRRs for concussion for a 10% increase in each neck strength predictor variable by multiplying the coefficients and confidence limits obtained from the model by ln(1.1) before exponentiating to obtain IRRs and CIs.[29] To allow comparison with other studies, we also present arithmetic means (SDs) by concussion and fit Poisson models for the untransformed neck strength data in order to obtain IRRs for the absolute change in neck strength variables (online supplemental results). We estimated the optimal cut point to predict concussion using the Youden Index[30] and calculated the true and false-positive rates using this cut-off. Model assumptions including normality of residuals for the mixed models were conducted using residual plots. Evidence for non-linearity was assessed by comparing models with restricted cubic splines to the linear model. The mean (0.13) and variance (0.12) for the Poisson model were similar with no evidence of dispersion. This was formally tested by fitting a negative binomial model and testing whether the overdispersion parameter differed from zero (p=0.32).

RESULTS

Demographics

A mixed international playing level existed within the study group. There were 23 current or previous senior international rugby players, 61 players had played at under-20 international level and 4 players whose international career had not progressed beyond under-18 level. One hundred and thirty-seven participants had played only domestic professional rugby. Player characteristics are presented in table 2.

Of the players recruited, 225 undertook testing at preseason (date: 24 August 2018 to 27 August 2018). Of these players, 179 were tested at mid-season (79.5%) (date: 26 January 2019 to 29 January 2019) and 74 at the end of season (32.8%) (date: 8 April 2019 to 11 April 2019). Twenty-two players did not attend mid-season testing because they had a current concussion, cervical or shoulder injury and 24 players did not report for testing at their scheduled time. At the end of season time point, 15 players did not report for testing due to a current concussion, cervical or shoulder injury and 136 players were absent from testing due to participation in an international rugby competition.

Changes in neck strength over time

When analysing changes in strength over time, all ranges demonstrated a significant increase in neck strength from preseason to mid-season time points (table 1). There was no significant

Table 1 Geometric mean neck strength (geometric SD) by range and time-point

					Percentage difference (95% CI)*		
Range	**Pre**	**Mid**	**Post**	**P value†**	**Mid vs pre**	**Post vs mid**	**Post vs pre**
Right rotation	24.6 (1.3)	27.4 (1.4)	26.3 (1.4)	<0.001	10.3% (6.8 to 13.9) p<0.001	−0.6% (-5.2 to 4.2) p=0.802	9.6% (4.6 to 14.8) p<0.001
Left rotation	25.2 (1.3)	27.0 (1.3)	25.8 (1.3)	<0.001	6.0% (2.8 to 9.4) p<0.001	−0.8% (-5.3 to 4.2) p=0.714	5.1% (0.5 to 9.9) p=0.029
Extension	41.9 (1.3)	44.2 (1.3)	41.7 (1.2)	<0.001	4.9% (2.3 to 7.2) p<0.001	−1.2% (-4.7 to 2.4) p=0.500	3.6% (0.0 to 7.3) p=0.049
Flexion	31.3 (1.3)	34.1 (1.3)	34.2 (1.3)	<0.001	8.5% (5.7 to 11.3) p<0.001	3.2% (-0.6 to 7.3) p=0.100	12.0% (7.9 to 16.3) p<0.001
Right flexion	24.4 (1.3)	25.6 (1.3)	24.7 (1.2)	<0.001	4.3% (1.6 to 7.0) p<0.001	1.1% (-2.6 to 5.0) p=0.561	5.5% (1.6 to 9.4) p=0.005
Left flexion	24.8 (1.3)	25.7 (1.3)	24.8 (1.2)	<0.001	3.4% (0.5 to 6.5) p=0.022	0.0% (-4.1 to 4.3) p=0.996	3.5% (-0.8 to 7.8) p=0.109
Total	173.5 (1.2)	185.6 (1.3)	178.7 (1.3)	<0.001	6.4% (4.1 to 8.8) p<0.001	0.2% (-3.0 to 3.6) P=0.883	6.7% (3.3% to 10.2) p<0.001
Flexion:extension	0.75 (1.3)	0.77 (1.2)	0.82 (1.3)	<0.001	2.7% (0.8 to 4.8) p=0.009	3.8% (0.9 to 6.8%) p=0.01	6.7% (3.8 to 9.7) p<0.001

*Significance level taken as p<0.017 for pairwise comparisons.
†P value from mixed effects regression model to take account of repeated measures data. Neck strength is log-transformed in the analysis.

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Table 2 Demographics by concussion

Variable	No concussion (n=196)	Concussion (n=29)	P value*
Age, mean (SD)	23.0 (4.0)	21.8 (4.0)	0.12
Height, mean (SD)	182.2 (6.13)	184.1 (6.68)	0.13
Weight, mean (SD)	96.7 (15.3)	96.8 (12.0)	0.96
BMI, mean (SD)	29.1 (4.1)	28.6 (3.2)	0.53
Years pro, median (IQR)	4(2–7)	3(2–4)	0.11
% with previous concussion (N)	52.1% (99)	65.5% (19')	0.18
Number of concussions	1 [0–2)	1 (0–3)	0.31
% international	40.5% (77)	58.6% (17)	0.07

*Two-sample t-test, Mann-Whitney U test, χ^2 test.
BMI, body mass index.

difference between mid-season and end-of-season for any range. Participants with postseason measures (n=72) tended to have lower neck strength values at mid-season than those who did not complete postseason testing (n=177). This may explain the disparity between the reported means between mid-season and end of season and the p value. Random effect parameters for the models are shown in online supplemental table 1.

Incidence of concussion

There were 30 concussions in 29 players recorded over the study period, giving an overall rate of 13.7 concussions per 1000 hours played. Nineteen concussions occurred between preseason and mid-season and 11 between mid and end of season time points. All concussions were recorded in match play.

Relationship between anthropometry, playing level and concussion incidence

No significant association was found between any anthropometric or playing level and concussion incidence (table 2).

Relationship between neck strength and concussion

There was a significant association between neck extension strength and the rate of concussion (p=0.044 unadjusted, p=0.019 covariate adjusted) (table 3). A 10% increase for extension strength was associated with a 13% decrease in concussion rate (p=0.019). There were no significant associations between concussion incidence and any other unique range or composite score, including the neck flexion:extension strength ratio. Our conclusions were the same when we obtained IRRs for the absolute change in neck strength (online supplemental table 2).

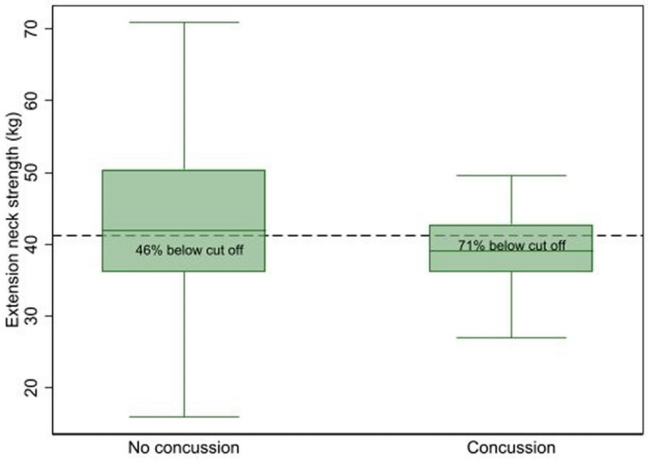

Figure 2 Box plot for neck strength—extension (kgf): Dashed line shows optimal cut-off point for concussion discrimination.

Identification of high-risk players according to neck extension strength

Players most at risk of sustaining a concussion had a neck extension strength score of 41 kg or below (71% true-positive and 46% false-positive rate) (figure 2). Fourty-six per cent of players lay below this threshold at baseline. Increasing neck strength from below to above the threshold would decrease the expected rate per 1000 player hours from 17.6 to 6.8 in these players (absolute rate difference=−10.8 (95% CI −20.4 to −1.2).

Table 3 Geometric mean baseline neck strength by concussion

Range	Players without concussion N=196	Players with concussion N=29	IRR* (95% CI) univariate	IRR† (95% CI) adjusted
Right rotation	24.6 (1.3)	24.7 (1.3)	1.00 (0.87 to 1.14) p=0.997	1.02 (0.87 to 1.20) p=0.77
Left rotation	25.2 (1.3)	25.0 (1.3)	1.01 (0.89 to 1.15) p=0.89	1.03 (0.94 to 1.12) p=0.55
Extension	42.5 (1.3)	38.2 (1.2)	0.87 (0.75 to 1.00) p=0.044	0.87 (0.78 to 0.98) p=0.019
Flexion	31.5 (1.3)	29.6 (1.3)	0.92 (0.79 to 1.06) p=0.23	0.94 (0.77 to 1.14) p=0.53
Right flexion	24.6 (1.3)	23.0 (1.2)	0.90 (0.78 to 1.04) p=0.14	0.91 (0.79 to 1.05) p=0.22
Left flexion	25.0 (1.3)	23.5 (1.2)	0.90 (0.78 to 1.04) p=0.16	0.92 (0.77 to 1.10) p=0.37
Total	174.7 (1.2)	166.0 (1.2)	0.92 (0.79 to 1.08) p=0.32	0.94 (0.80 to 1.12) p=0.50
Flexion:extension	0.74 (1.2)	0.78 (1.2)	1.13 (0.93 to 1.38) p=0.23	1.13 (0.97 to 1.33) p=0.13

*Incidence rate ratio for a 10% increase in each variable.
†Adjusted for age, club, BMI.
BMI, body mass index.

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Inter-reliability of raters

Inter-rater reliability was measured using the intraclass correlation coefficient, comparing the variability of ratings of the same individual to the total variation of ratings for all individuals. Measurements are averaged over the three trials. The overall agreement was moderate to excellent for all ranges of neck strength measures between two raters (0.707–0.985)

DISCUSSION

We hypothesised an association between reduced neck muscle strength and increased concussion incidence in male professional rugby players. This study is the first to identify a specific neck strength range associated with increased concussion rate in an athletic population and the first to identify that reduced neck extension strength is a risk factor for concussion in male professional rugby players. We observed a rate of 13.7 concussions per 1000 player-match-hours, which is consistent with concussion rates in professional male Rugby Union.[5 31 32] When adjusted for player match exposure, our results demonstrate that for every 10% increase in extension strength, there is a 13% decrease in concussion rate. Furthermore, we identify what might be described as a minimally acceptable neck extension strength of 41 kg. By highlighting those athletes with neck extension strength below this range, one can identify 68% of players who will sustain a concussion over the course of a professional rugby season. As stated, the specificity of this finding is not high, highlighting the importance of a team wide neck extension strengthening programme, with interventions not limited to players that fall below this cut-off. This is especially important considering that concussion rates reduce exponentially with increasing neck extension strength.

Our results build on earlier work by Collins *et al*[18] who found that a lower composite neck strength score was a risk for concussion in a mixed group of high-school athletes. While some studies have found no significant association between neck muscle function and head movement velocity during simulated impacts,[13 33 34] many of the lab-based studies have simulated impacts using a sudden backward pull of a subject's head,[33–38] provoking a reaction from the flexor muscle group of the neck. Our results suggest that the extensor muscles may have a larger role to play than previously thought in attenuating forces of impact. The cervical extensors are consistently reported to be the muscle group that generate the highest isometric neck force.[25 39 40] It may be possible, therefore, that this strength range represents the greatest defensive mechanism at reducing the force of impacts during sagittal plane impact, previously identified as the most common direction of concussive impacts in male professional rugby players.[12]

We found no significant association with concussion and any anthropometric or playing measure including age, height, weight, BMI, concussion history, career length or international playing level. This is consistent with the previous literature.[18 41] There was a trend towards a greater risk of concussion in those players who have played rugby at senior or junior international levels, but it is speculative as to the reasons. It may be that international players find themselves in contact situations more regularly or they have a higher propensity for risk-taking behaviour, both of which may make them better players and, therefore, more likely to be selected at international level. The conclusions are supported in a recent review by Burger *et al* who found that concussive injuries are most likely to occur in rugby players playing at a higher level of competition.[42]

It is our belief that this study is also the first to track professional rugby players' neck strength over the course of a professional season using three equally spaced time points. A significantly lower neck strength existed among the study population at preseason compared with mid-season and end of season time points correlating with the concussion incidence in this study. Nineteen of the 30 concussions occurred between pre-season and mid-season, when average player neck strength was lower, 11 concussions were sustained between the mid and end of season time points. This suggests that sporting organisations may benefit from a strength and conditioning focusing on neck strength interventions at preseason to mitigate the risk of concussions in the first half of the season. Research has shown that it is possible to make significant improvements in extensor neck strength through targeted interventions.[34 43 44]

These insights provide important new information, particularly to rugby playing populations and their support teams who are able to direct targeted intervention(s) to minimise risk. Rugby has one of the highest incidence rates of concussion across all sports[45 46] and injury data show no sign of this incidence reducing.[6] Although this is due to a multitude of different factors, including better identification and reporting, similar trends are observed in a number of contact sports.[47 48] Significant financial[49 50] and future health concerns have also been linked to head impacts,[51 52] although it must be emphasised that the link between concussion and future health concerns remains unproven.

The evidence presented in this investigation highlights the importance of further research including evaluating the presence of other modifiable risk factors, and prospective interventional studies investigating the efficacy of neck strength training in reducing concussion incidence, across different groups (age, gender and playing level). Other physical qualities of neck function such as joint proprioception, muscle stiffness, range of movement and even vestibular and oculomotor function have been shown to be factors in other common injuries[53 54] and should be considered as part of future research in concussion risk reduction.

We feel that the methods employed in this study were robust. The method of strength assessment used[28] has been shown to have good to high inter-session and between-session reliability and our reliability data between testers demonstrated high kappa coefficients. The inclusion of player match exposure was an important factor in determining the true concussion rate, as those players who had low playing time over the course of the season were accounted for in the final statistical analysis. We were also fortunate to conduct this study on a well-controlled population whose primary focus was their participation in a single sport. All clinicians responsible for diagnosis and reporting concussions during the study period were blinded to the neck strength results of this study and were, therefore, less likely to be open to bias when making clinical decisions on players' diagnoses.

Unfortunately, we had a significant drop out rate at end of season testing due to an international rugby competition. This made our end of season strength measures appear low when compared with mid-season, however, this was accounted for when analysing the difference in neck strength between time points. The difference in mean strength observed, occurred because players who completed postseason testing had lower mid-season scores than the average and, therefore, were over-represented in the end of season mean. Although we studied a homogeneous group of professional rugby players, they all competed in the same league, where playing style and training methods may be specific to this region, potentially limiting translation to professional leagues in other nations (ie, Premiership, England; Top 14, France; Super Rugby, New Zealand and Australia). Finally, our investigation was conducted on professional male players. Therefore, our conclusions may not be

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applicable to professional female players, and amateur and youth male players.

CONCLUSION

We demonstrate that low isometric neck extension strength is a risk factor for concussion in male professional rugby players and have identified a 'minimally acceptable' neck extension range that may help to guide sports medicine departments in formulating preseason conditioning programmes. Future research should examine if neck strengthening interventions during the early season reduces concussion incidence in season.

Key messages

What is already known on this topic?

- Poor neck strength in rugby players is thought to increase the risk of concussion during match play. Only one published research paper currently exists that demonstrates the association between poor neck strength and concussion incidence and no rugby players were included in this study population.

What this study adds?

- The findings of this study demonstrate that poor neck extension strength is associated with an increase in concussion rates in male professional rugby players, with a 13% reduction in concussion rate for every 10% increase in neck extension strength.
- How this study might affect research, practice or policy:
- Sporting organisations may benefit from a strength and conditioning regime, including neck extension strengthening throughout the course of a rugby season, with increased neck extension strength exponentially correlated with a reduction in concussion risk.

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Collaborators There were no collaborating groups involved in this study.

Contributors TF conceptualised the study methods and along with EB, undertook data collection. TF wrote the original draft of the manuscript. All authors analysed the data and supported the construction of the manuscript. All authors read and approved the final copy of the manuscript. TF is the guarantor

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Provenance and peer review Not commissioned; externally peer reviewed.

Data are not publically available, however, access can be provided following suitable request and approval by the sporting union.

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